CUSTOMER CHURN ANALYSIS

Executive findings, customer-risk signals and retention priorities
Python • Pandas • Seaborn • Matplotlib

Purpose

Translate exploratory analysis into a decision-oriented view of where churn is concentrated, what to investigate next, and how a retention team could test targeted actions.

CUSTOMERS ANALYSED	CHURNED	OVERALL CHURN
7,043	1,869	26.54%
21 columns in dataset	5,174 retained	About 1 in 4 customers

Executive summary

• **1. Retention risk is material.** The notebook reports a 26.54% churn rate across 7,043 customers.

• **2. Early tenure is the clearest priority signal.** The 0–6 month segment shows 52.94% churn, falling to 9.51% among customers with 49–72 months of tenure.

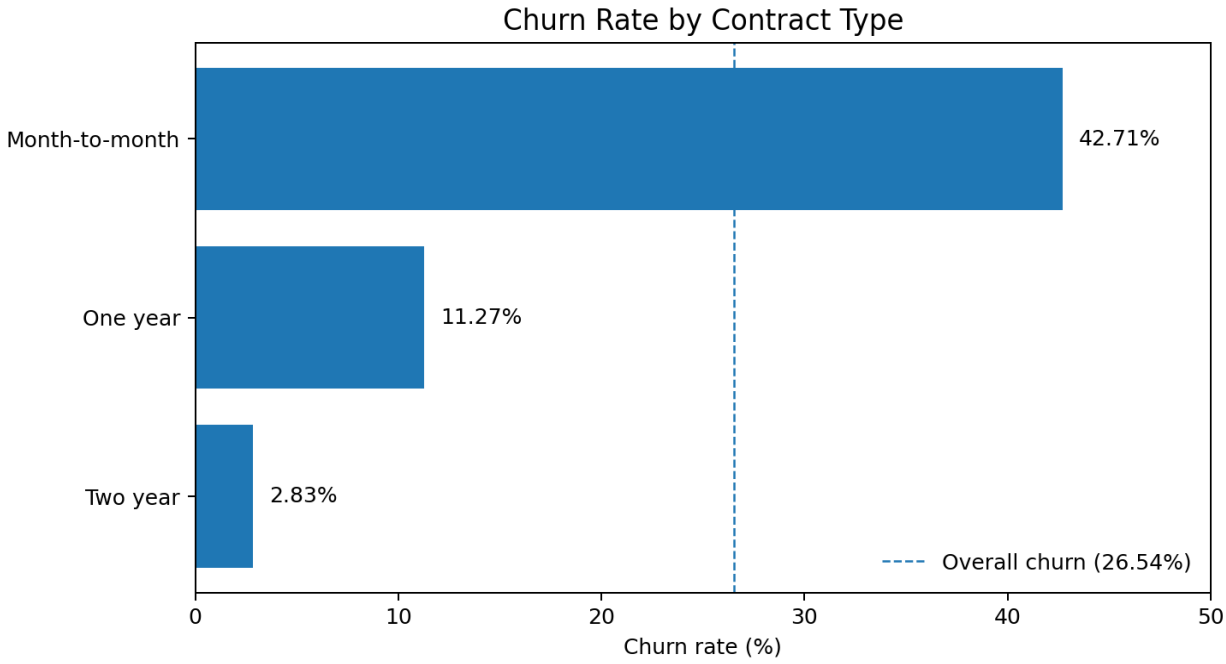
• **3. Contract and payment behaviour separate risk groups.** Month-to-month customers show 42.71% churn; electronic-check customers show 45.29%.

• **4. Service experience warrants investigation.** Fiber-optic customers and customers without selected support/security services have churn rates around 39–42%.

• **5. Treat these as signals, not causes.** This is descriptive EDA. Validate drivers with deeper analysis and controlled retention experiments before allocating major spend.

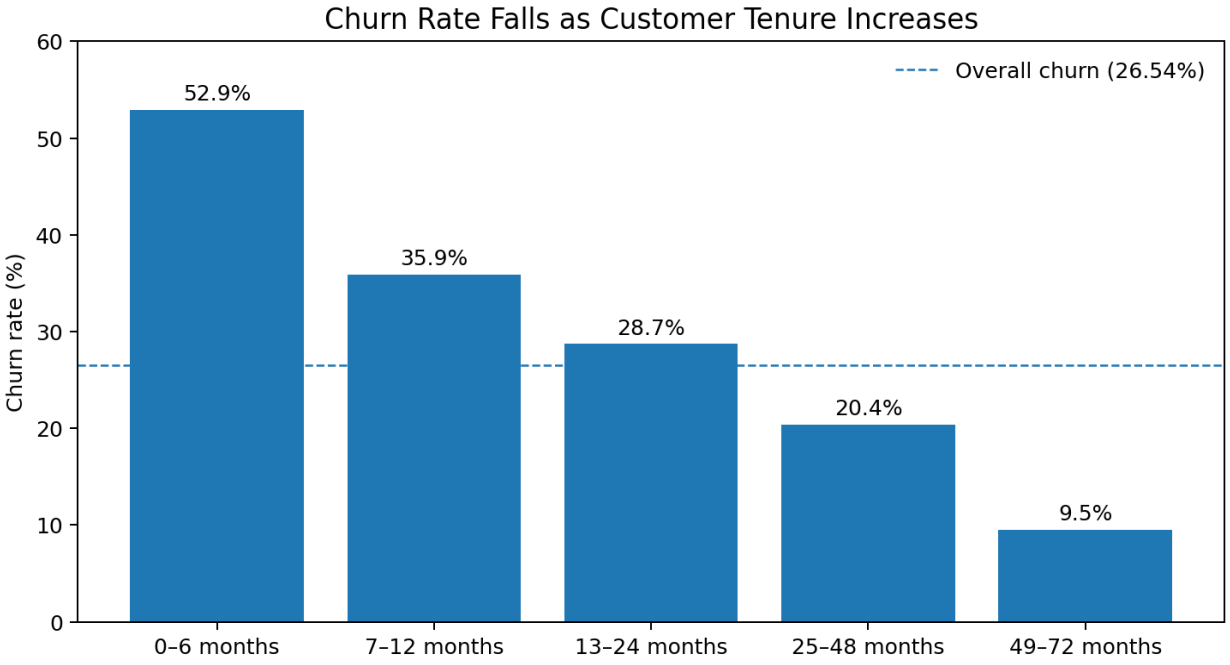
1. Churn distribution

The overall rate provides a baseline for comparing customer segments. The segment comparisons below use within-group churn rates, which are more decision-useful than raw churn counts alone.



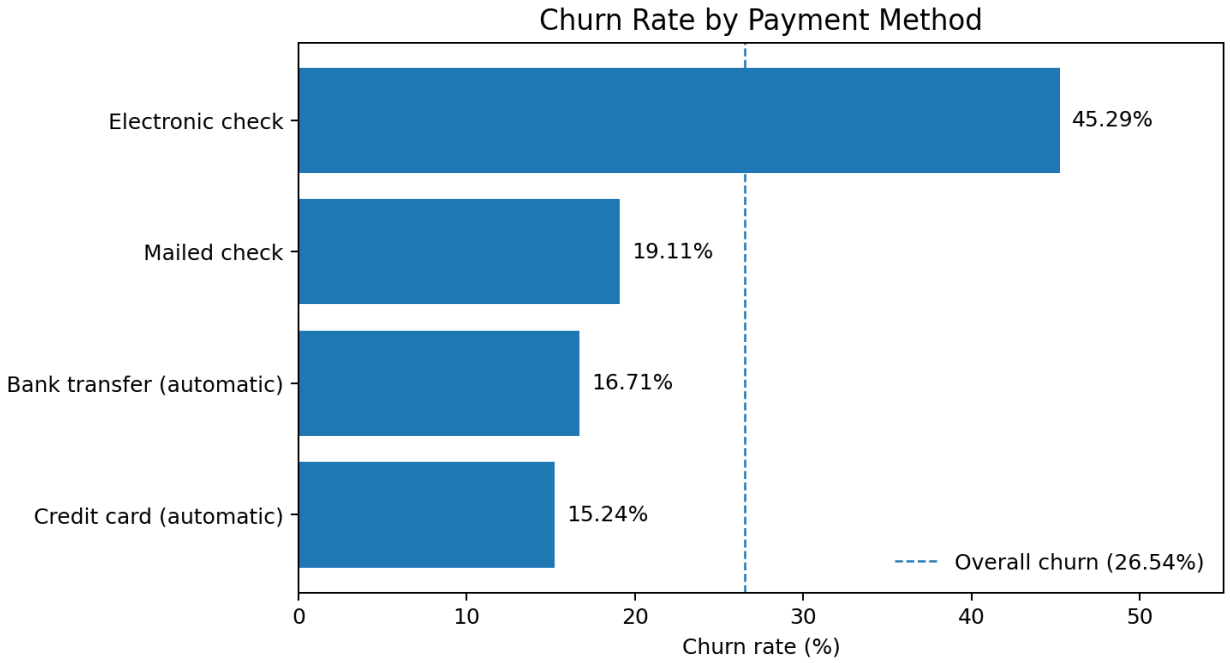
Contract type is strongly associated with observed churn: month-to-month customers have a 42.71% churn rate, versus 2.83% for two-year contracts. This may reflect commitment, customer tenure, pricing, or underlying satisfaction; the EDA cannot separate these explanations.

2. Early-tenure retention



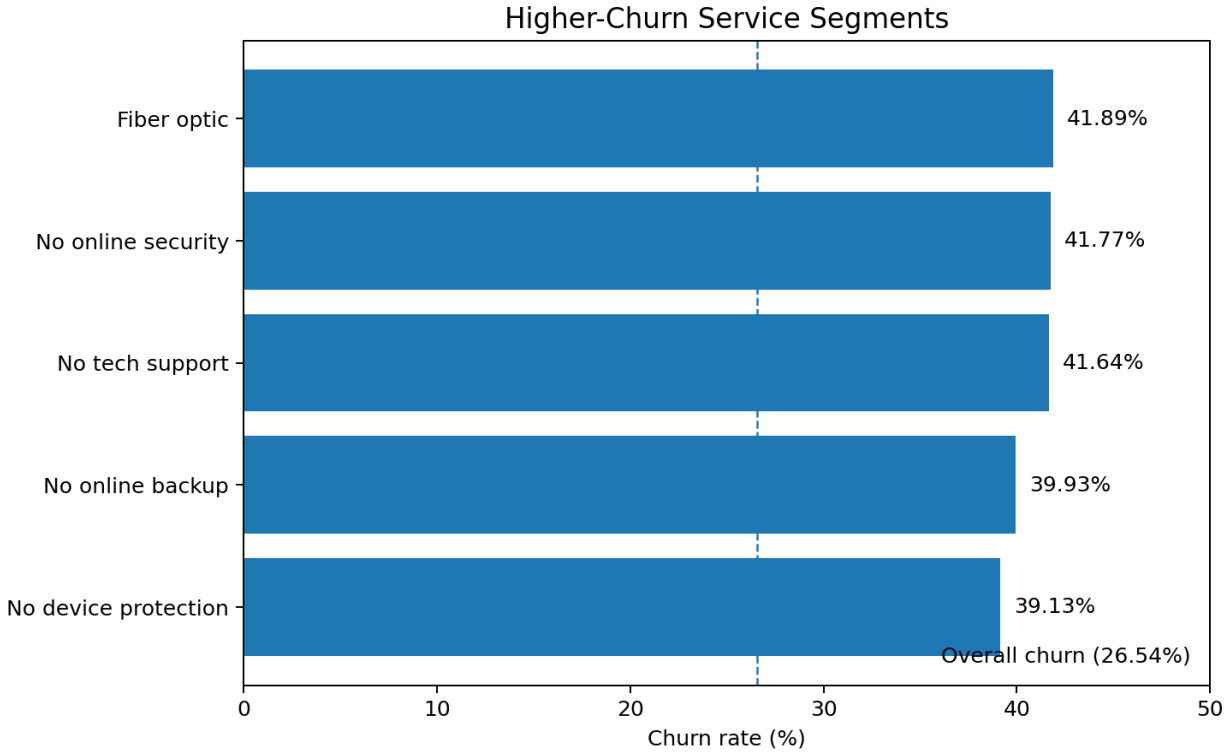
The highest observed churn is in the first six months (52.94%). The monotonic decline across tenure bands suggests that early-life customer experience should be investigated first. Recommended analysis: cohort-level churn by signup month, early support contacts, service outages, installation issues and first-bill complaints.

3. Payment and billing



Electronic-check customers have a 45.29% churn rate, materially above the 26.54% portfolio baseline. Paperless-billing customers also show higher observed churn (33.57% vs 16.33% for non-paperless billing). These associations could reflect customer mix or product/plan differences; investigate payment friction and billing experience before interpreting them as payment-method effects.

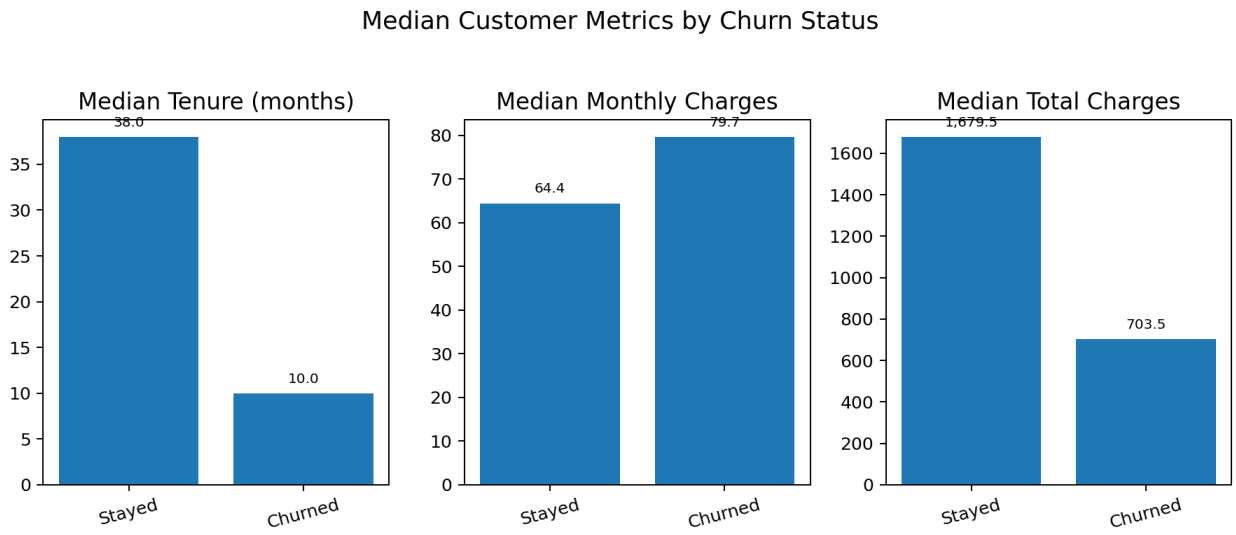
4. Service and customer-profile signals



Fiber-optic service is associated with 41.89% churn. Customers without online security (41.77%) and without tech support (41.64%) also show elevated churn. A useful next step is to join service-level churn with complaints, outage/quality measures, support tickets, plan price and local availability.

Profile differences from the notebook include 41.68% churn among senior citizens vs 23.61% among non-senior customers, and 32.96% among customers without a partner vs 19.66% among customers with a partner. These should guide careful segmentation and service research, not discriminatory treatment or assumptions about individuals.

5. Customer economics



Churned customers have a median tenure of 10 months versus 38 months for retained customers. Their median monthly charges are higher (79.65 vs 64.43), while median total charges are lower (703.55 vs 1,679.52), consistent with shorter tenure. A useful follow-up is to compare charge levels within tenure, contract and service groups to avoid conflating price with customer lifecycle.

6. Recommended action plan

Priority	Action	Measurement
P1 — First 6 months	Test onboarding, proactive check-ins and rapid issue resolution for new customers.	30/90/180-day churn; complaints; service contacts
P1 — Contract	Analyse cancellation reasons for month-to-month customers; test value-led plan options.	Churn by contract/cohort; conversion; margin
P1 — Payment	Review electronic-check payment journey, failed payments and billing confusion.	Payment failure rate; churn; customer effort
P2 — Fiber/service	Compare network quality, outages, price and support cases for fiber customers.	Churn by quality/price/support segment
P2 — Support/security	Identify unmet support needs and test helpful, relevant service education.	Take-up; resolution; churn; customer satisfaction
P3 — Measurement	Run controlled tests or phased rollouts with a holdout group where feasible.	Incremental churn reduction and cost per retained customer

7. Methodology, caveats and next steps

Preparation recorded in notebook: 7,043 rows × 21 columns; no duplicate customer IDs; 11 blank/unparseable TotalCharges values, all recorded for zero-tenure customers, converted to zero after numeric conversion.

Method: descriptive cross-tabulations and visual comparisons by churn status, contract, tenure band, service, payment method and charges. Rates are calculated within each category.

Limitations: no predictive model is trained; no causal inference; segment differences may be confounded; results depend on the exact CSV version. The figures in this report are taken from the uploaded notebook's saved outputs and should be refreshed by re-running against the intended dataset.

Next analytical phase: validate data and definitions, investigate cohort and service-ticket drivers, then build a separate baseline churn model. Use a stratified train/test split, leakage checks and metrics such as recall, precision, ROC-AUC

and calibration. Link model thresholds to the cost of false positives and missed churners.

Analyst conclusion

The most actionable starting point is early-tenure retention, followed by month-to-month contract and electronic-check segments. Treat these as prioritisation hypotheses, then verify the underlying customer experience and measure intervention impact with controlled tests.

Source: uploaded Customer_Churn_Analysis.ipynb saved outputs. Currency units are not specified in the notebook, so charge metrics are reported as dataset values without a currency symbol.